

## AHA SCIENTIFIC STATEMENT

# Heart Failure Occurring in the Perinatal Period: A Scientific Statement From the American Heart Association

Demilade A. Adedinsowo, MD, MPH, Chair; Haywood L. Brown, MD; Afshan B. Hameed, MD, MBA; Dennis McNamara, MD; Mulubrhan F. Mogos, PhD, MSc, FAHA; Jenna Skowronski, MD; Arthur Vaught, MD; Marie-Louise Meng, MD; Modele O. Ogunniyi, MD, MPH, FAHA; Nosheen Reza, MD, MS, Vice Chair; on behalf of the American Heart Association Women's Health Science Committee of the Council on Clinical Cardiology; Council on Cardiovascular Surgery and Anesthesia; and Council on Cardiovascular and Stroke Nursing

**ABSTRACT:** Heart failure in the perinatal period remains ambiguous in definition and management despite its recognition as a unique disease state. The true incidence and prevalence of heart failure or left ventricular systolic dysfunction during pregnancy and the postpartum period are unknown, although a prevalence as high as 1% to 2% has been reported in the general adult US population. Assessment of heart failure can be challenging in the pregnant or postpartum state, during which symptoms affecting physical function (eg, dyspnea, exercise intolerance, fatigue, and lower-extremity edema) are prevalent because of physiological changes. Delays in the recognition and diagnosis of heart failure during the perinatal period contribute to adverse maternal outcomes, highlighting the need for evidence-based definitions and thresholds, improved diagnostic criteria to aid disease recognition, and effective screening tools. This scientific statement focuses on heart failure with reduced and mildly reduced ejection fraction in the context of pregnancy and the postpartum period, caused by various forms of cardiomyopathy. It addresses challenges related to recognizing heart failure in obstetric patients, outlines established treatment standards, and underscores potential areas for research. To improve the management of preexisting and de novo heart failure in obstetric patients, standardization of disease definitions, specific therapeutic options, implementation of effective screening tools, and interventions to improve maternal health equity are imperative. Future directions include prioritizing the inclusion of pregnant and postpartum individuals in heart failure studies, implementing interventions that facilitate early disease detection, and ensuring the timely initiation of appropriate therapies with the goal of reducing adverse outcomes associated with perinatal heart failure.

**Key Words:** AHA Scientific Statements ■ cardiomyopathies ■ heart failure, systolic ■ postpartum period ■ pregnancy ■ ventricular dysfunction, left

In the United States, cardiovascular disease (CVD) remains a leading cause of maternal morbidity and mortality, accounting for 27% of pregnancy-related deaths in 2023,<sup>1,2</sup> a trend observed in many other developed countries.<sup>3</sup> Pregnant women with heart failure (HF) have a markedly elevated risk, with mortality rates reported to be 32 times higher than in those without HF at the time of delivery according to data from the Nationwide Inpatient Sample.<sup>4</sup> It is important to note that >80% of maternal deaths attributed to CVD are considered preventable,<sup>5</sup> underscoring the need for improved

recognition, management, and prevention strategies in this population.

Cardiomyopathies in pregnancy account for a substantial portion of cardiovascular conditions in pregnancy, with dilated cardiomyopathy and peripartum cardiomyopathy (PPCM) accounting for ≈92% of all reported cases.<sup>6</sup> In 2023, cardiomyopathies were responsible for 10.2% of all pregnancy-related deaths, and hypertensive disorders of pregnancy accounted for 6.4%.<sup>1</sup> Compared with other cardiovascular conditions during pregnancy, cardiomyopathy had the strongest association with the

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development of major adverse cardiac events.<sup>7</sup> HF is one of the most prevalent complications among pregnant and postpartum women with preexisting heart disease, affecting 6% to 33% of this population.<sup>8,9</sup> Cardiomyopathy and HF pose a substantial risk to the health of a mother and her offspring. Maternal risks include arrhythmia, stroke, worsening cardiac function, preterm delivery,<sup>10</sup> caesarean delivery, postpartum hemorrhage, mortality,<sup>11</sup> poor mental health, and poor quality of life.<sup>12,13</sup> Fetal and neonatal risks include fetal growth restriction, prematurity, small-for-gestational-age baby, prolonged intensive care unit stay, stillbirth, and neonatal mortality.<sup>8</sup>

In this scientific statement, we focus on HF with a low ejection fraction that is present, worsening, or occurring de novo during pregnancy and within 12 months postpartum. In addition to PPCM and idiopathic dilated cardiomyopathy, these include various forms of cardiomyopathy such as endocrine, genetic, inflammatory, infectious, ischemic, nutritional, stress induced, tachycardia mediated, and toxin induced, including chemotherapy. The rationale for this combined approach stems from the challenge in distinguishing preexisting left ventricular dysfunction from dysfunction arising de novo when no baseline testing is available before pregnancy. In addition, clinical evaluation and management of left ventricular dysfunction—whether preexisting, worsening, or new onset—are guided by a unified care pathway, with additional management tailored to the underlying cause when identified.

For the purposes of this document, we define the perinatal period as extending from the onset of pregnancy through 12 months postpartum.<sup>14,15</sup> In addition, we synthesize and critically evaluate current evidence from recent peer-reviewed literature and authoritative clinical guidelines. Our appraisal includes the 2022 American Heart Association/American College of Cardiology/Heart Failure Society of America guidelines for managing heart failure,<sup>15a</sup> specifically the section addressing HF and pregnancy, and the 2025 European Society of Cardiology guidelines for the management of CVD during pregnancy.<sup>15b</sup> This scientific statement also encompasses conditions included within the most recent universal definition for HF with reduced and mildly reduced ejection fraction.<sup>16</sup> HF with preserved ejection fraction in the perinatal period<sup>17</sup> is beyond the scope of this document.

## DEFINING HF IN THE PREGNANT AND POSTPARTUM STATE

### Definition

HF in the context of pregnancy and the postpartum period remains ambiguously defined. Although it is recognized that assessing and managing HF occurring before pregnancy, during pregnancy, and postpartum

should be individualized,<sup>18</sup> scientific evidence for clinical guideline development remains limited, and existing recommendations are narrow in scope. Historically, the concept of HF occurring in the perinatal period has been centered around PPCM, a diagnosis of exclusion, with its definition restricted to a left ventricular ejection fraction (LVEF) cutoff value of <45%. Otherwise, HF that occurs during the perinatal period in the context of preexisting structural heart disease is assessed within relevant disease-specific frameworks. The LVEF threshold of <45% for PPCM was introduced by the National Heart, Lung, and Blood Institute in 1992.<sup>19</sup> Subsequently, Hibbard et al<sup>20</sup> recommended strict echocardiographic criteria analogous to those for idiopathic dilated cardiomyopathy. However, the 1997 National Heart, Lung, and Blood Institute workshop revised the PPCM criteria without including an LVEF cutoff.<sup>21</sup> The European Society of Cardiology updated the definition in 2010,<sup>22</sup> deeming the previous time frame and cutoff values arbitrary. In 2019, the European Society of Cardiology acknowledged that women with LVEF between 45% and 50% may be diagnosed with PPCM.<sup>23</sup> The various definitions established over time are summarized in Table 1. In contrast to the time- and LVEF-based definition of PPCM, HF is universally defined as “a clinical syndrome with symptoms and or signs caused by a structural and or functional cardiac abnormality with elevated natriuretic peptides or other objective evidence of cardiogenic pulmonary or systemic congestion.”<sup>16</sup> This definition offers 4 classifications based on LVEF values: (1) HF with reduced ejection fraction, LVEF ≤40%; (2) HF with mildly reduced ejection fraction, LVEF 41% to 49%; (3) HF with preserved ejection fraction, LVEF ≥50%; and (4) HF with improved ejection fraction, baseline LVEF of ≤40%, a ≥10-point increase from baseline LVEF, and a second measurement of LVEF of >40%. This proposed classification by LVEF value was intended to facilitate treatment decision-making in relation to guideline-directed medical therapy. There are limited data on the application of these LVEF-based classifications to HF occurring during the perinatal period. Applying current guidelines to define HF in pregnant and postpartum women is limited by several factors. These include sex-based variations in LVEF classification thresholds,<sup>24</sup> pregnancy-related myocardial structural changes,<sup>3</sup> a lack of validated reference standards for cardiac chamber sizes and biomarkers in pregnancy, and insufficient clinical trials on medical therapy for HF in this population.

In this document, we focus on HF with LVEF <50% (encompassing HF with reduced ejection fraction and HF with mildly reduced ejection fraction) and acknowledge that identification and recognition during the perinatal period require careful consideration of normal pregnancy physiology and integration with objective cardiac assessments.

**Table 1. Evolution of PPCM: Definitions and Diagnostic Criteria**

|                                                                              | Description and time frame                                                                                                           | Description of left ventricular dysfunction                                                                                                                                                                                                            | Additional characteristics                                                                                                         |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| NHLBI Idiopathic DCM Workshop criteria, <sup>19</sup> 1992                   | Development of left ventricular dilatation and systolic dysfunction in the last month of pregnancy or 5 mo postpartum                | Idiopathic DCM was defined as ejection fraction by echocardiography, radionuclide scanning or ventriculography <45% or M-mode fractional shortening <30%, or both in the presence of an end-diastolic left ventricular diameter >2.7 cm/m <sup>2</sup> | The absence of a definable cause or demonstrable heart disease during pregnancy<br>Considered a separate subtype of idiopathic DCM |
| NHLBI and Office of Rare Diseases PPCM Workshop criteria, <sup>21</sup> 1997 | HF in the last month of pregnancy or within 5 mo of delivery                                                                         | LVSD demonstrated by classic echocardiographic criteria (depressed shortening fraction or ejection fraction)                                                                                                                                           | Absence of an identifiable cause of HF<br>Absence of recognizable heart disease before the last month of pregnancy                 |
| Hibbard et al, <sup>20</sup> 1999                                            | HF within the last month of pregnancy or 5 mo postpartum                                                                             | Strict echocardiographic criteria of left ventricular dysfunction: ejection fraction <45%, M-mode fractional shortening <30%, or both and end-diastolic dimension >2.7 cm/m <sup>2</sup>                                                               | Absence of preexisting heart disease<br>No determinable cause                                                                      |
| ESC simplified PPCM definition, <sup>22</sup> 2010                           | Idiopathic cardiomyopathy presenting with HF secondary to LVSD occurring toward the end of pregnancy or in the months after delivery | Left ventricle may not be dilated, but ejection fraction is nearly always reduced to <45%                                                                                                                                                              | No other cause of HF is found<br>A diagnosis of exclusion                                                                          |
| ESC PPCM definition, <sup>23</sup> 2019                                      | HF secondary to LVSD; occurrence toward the end of pregnancy or in the months after delivery (mostly in the month after delivery)    | LVEF <45%; patients with an ejection fraction between 45% and 50% may occasionally be diagnosed with PPCM                                                                                                                                              | No other identifiable cause of HF                                                                                                  |

DCM indicates dilated cardiomyopathy; ESC, European Society of Cardiology; HF, heart failure; LVEF, left ventricular ejection fraction; LVSD, left ventricular systolic dysfunction; NHLBI, National Heart, Lung, and Blood Institute; and PPCM, peripartum cardiomyopathy.

## Symptom Evaluation

Symptom assessment in relation to HF classification remains challenging in pregnancy because of the overlap with expected symptoms arising from physiological changes.<sup>25,26</sup> In addition, the effects of pregnancy on physical function are not objectively defined and can vary from person to person. Many of the symptoms that suggest the presence of HF such as dyspnea, exercise intolerance, fatigue, and ankle swelling/edema can be observed with normal pregnancy,<sup>16,26–35</sup> complicating recognition of the HF state. Although the presence of orthopnea warrants prompt evaluation for cardiac disease because of its association with cardiac pathology, some studies have shown that pregnant patients with normal left ventricular systolic function may experience orthopnea or, more frequently, supine intolerance, characterized by desaturation or other symptoms when lying flat, mimicking orthopnea.<sup>31,36,37</sup> This makes it challenging to risk stratify patients with symptom-based criteria alone. Nevertheless, self-reported cardiac symptoms during pregnancy should be taken seriously and should be evaluated, particularly in relation to existing risk factors and clinical signs.<sup>33</sup> The severity of physical limitations attributed to HF is most frequently assessed with the New York Heart Association classification system, a clinician-assigned subjective assessment of symptoms and functional capacity among patients with symptomatic HF. It is used in clinical practice to determine eligibility for specific HF treatments and to assess disease severity and prognosis.<sup>15a</sup> However, the New York Heart Association classification metric has been demonstrated

in nonpregnant individuals to be a poor discriminator of functional impairment.<sup>39</sup> This suggests that its utility in pregnancy may be more limited, although research in this area is lacking. Nevertheless, studies in pregnant cohorts have shown an association between New York Heart Association class III/IV symptoms and adverse cardiac events.<sup>9,15b</sup> Patient-reported outcome measures such as the Kansas City Cardiomyopathy Questionnaire<sup>41</sup> and the Minnesota Living with Heart Failure Questionnaire<sup>42</sup> are validated instruments used to assess HF symptom severity and quality of life in clinical care and have been evaluated as important end points in cardiovascular clinical trials. A small study among obstetric patients showed a linear association between increasing symptom severity (indicated by lower scores), assessed with a pregnancy-adapted version of the Kansas City Cardiomyopathy Questionnaire, and worsening degree of left ventricular dysfunction,<sup>31</sup> suggesting that patient-reported outcome measures may offer improved assessment of symptoms and cardiac function during the perinatal period.

## Assessment of Cardiac Function and Cardiogenic Pulmonary or Systemic Congestion

### Electrocardiography

A 12-lead ECG is considered one of the primary diagnostic tests in the evaluation of HF for all adult patients.<sup>15a</sup> It provides information on heart rate, rhythm, QRS morphology, and cardiac interval duration, which may inform HF cause and prognosis.<sup>15a</sup> It is important to note that

changes on the ECG, including increased heart rate, p-wave duration, and QTc interval, may occur with normal pregnancy, but most remain within normal limits.<sup>43,44</sup> Changes in P, QRS, and T-wave morphology and axes have also been reported,<sup>15b,45–47</sup> possibly resulting from hormonal and or structural adaptations to pregnancy as demonstrated on echocardiography.<sup>48</sup>

### **Echocardiography**

There is substantial evidence that LVEF values vary by sex, with women tending toward higher normal values compared with men.<sup>24,49–51</sup> During normal pregnancy, LVEF is reportedly unchanged from prepregnancy values and remains mostly within the normal limits (using thresholds derived in nonpregnant adults) based on a series of independent studies.<sup>3,47,52</sup> However, a mild decrease in LVEF has been described in the second trimester persisting through the early postpartum period.<sup>3,52–54</sup> Echocardiographic changes in cardiac structure (notably affecting left ventricular mass, wall thickness, dimensions, and filling pressures, as well as left atrial volumes)<sup>3,47,55,56</sup> and function (affecting cardiac output, contractility, and global longitudinal systolic strain) are known to occur with normal pregnancy.<sup>53,54</sup> Although multiple individual and relatively small studies have described echocardiographic changes in pregnancy,<sup>57</sup> normal values and appropriate reference ranges for cardiac chamber size and function from 2-dimensional imaging in pregnant individuals across each trimester and in postpartum individuals have yet to be established by cardiovascular imaging societies.<sup>57a</sup>

### **Magnetic Resonance Imaging**

Cardiac magnetic resonance imaging without contrast is safe during pregnancy and is used mostly in the evaluation of pregnant individuals with preexisting structural heart disease.<sup>58</sup> It may also be considered if echocardiographic assessment is limited or inconclusive. The use of gadolinium during pregnancy has potential fetal risk. Thus, it is advisable that gadolinium be administered solely in circumstances in which the potential benefits clearly outweigh the risks.<sup>59,60</sup> As with echocardiography, normal values for cardiac magnetic resonance imaging–based assessment of cardiac structure and function in pregnant and postpartum individuals have not been established.

### **Chest Radiography**

Although chest radiography can be used to identify evidence of cardiogenic pulmonary congestion, non-radiation-based imaging modalities are preferred during pregnancy to minimize the risk of fetal exposure.<sup>60</sup> This test should be performed only in symptomatic pregnant women when no other diagnostic methods are available or when existing methods do not adequately explain symptoms.<sup>15b</sup> Imaging protocols should be adapted to minimize radiation exposure.<sup>60</sup> However, the fetal radiation dose from chest radiography is estimated to range from 0.0005 to 0.01 mGy, varying by gestational age.

This exposure is classified as low and poses no significant risk to the fetus.<sup>59</sup>

### **Serum Biomarkers**

BNP (B-type natriuretic peptide) and NT-proBNP (N-terminal pro-b-type natriuretic peptide) levels generally remain within normal limits or become mildly elevated in normal pregnancy, with a greater degree of elevation in early pregnancy compared with the third trimester.<sup>61–64</sup> Although studies have shown that natriuretic peptides demonstrate high negative predictive value in obstetric patients,<sup>65</sup> their positive predictive value is comparatively lower than in nonpregnant individuals.<sup>63</sup> Elevated NT-proBNP values (>125 pg/mL) have been reported in pregnant women, particularly in the first trimester, compared with nonpregnant women,<sup>64</sup> underscoring the importance of considering these variations in the interpretation of NT-proBNP results in pregnant individuals. The development of acute HF in pregnancy and postpartum is characterized by markedly elevated natriuretic peptide levels compared with those without HF.<sup>63</sup> Although maternal obesity and preeclampsia can alter natriuretic peptide levels, BNP values <100 pg/mL and NT-proBNP values <128 pg/mL retain a high negative predictive value of 100% and >95%, respectively, for excluding HF on the basis of limited data.<sup>63</sup> In the absence of robust data, an expert panel convened by the American College of Cardiology Cardio-Obstetrics Work Group proposed abnormality threshold values during pregnancy.<sup>63</sup> Emerging studies have demonstrated potential clinical relevance at these suggested thresholds.<sup>66–68</sup>

## **EPIDEMIOLOGY AND BURDEN OF HF WITH LEFT VENTRICULAR SYSTOLIC DYSFUNCTION IN THE PERINATAL PERIOD**

The prevalence of left ventricular systolic dysfunction (LVSD) in the general population is estimated to range from 1% to 2%.<sup>69–71</sup> Among pregnant and postpartum individuals, the true incidence and prevalence of any LVSD are unknown. Epidemiological estimates often focus on specific disease phenotypes such as PPCM. However, it is believed that many patients with perinatal LVSD likely go undiagnosed, particularly those who are asymptomatic or exhibit milder symptoms.<sup>72</sup> More recently, systematic screening studies in obstetric cohorts have estimated the prevalence of any LVSD to be as high as 4%<sup>73</sup> (LVEF <50%) and 2.2%<sup>74</sup> (LVEF <55%) in Nigeria and Pakistan, respectively.

### **Preexisting HF in the Perinatal Period**

Preexisting HF in the general population of apparently healthy pregnant and postpartum women is reportedly



uncommon because of limited data in this population; however, it poses a significant risk to both the mother and the baby. Within ROPAC (Registry of Pregnancy and Cardiac Disease), a prospective registry of pregnancy experiences and outcomes among women with heart disease enrolled from 53 countries, the overall prevalence of preexisting LVSD (LVEF <40%) was 5.3%. Within the subgroup of women with ischemic heart disease, the prevalence was as high as 12.8%.<sup>75</sup> Many studies reporting HF as a clinical outcome in pregnant and postpartum women frequently do not differentiate between preexisting and de novo HF in the perinatal period. In addition, it is not considered standard practice for women of reproductive age to routinely undergo systematic screening or surveillance for HF or cardiomyopathy before conception.<sup>28</sup> The limited literature on this subject precludes a comprehensive evaluation of preconception cardiovascular health, particularly HF, among women of reproductive potential.

### De Novo HF and Cardiomyopathy Types

PPCM is the most common form of de novo HF diagnosed during pregnancy and postpartum.<sup>6,76</sup> However, there is insufficient evidence from retrospective studies that all obstetric patients diagnosed with PPCM underwent comprehensive evaluations to exclude all other potential causes of LVSD. Although the true incidence and prevalence of PPCM remain debated, published estimates vary substantially across racial and ethnic groups and geographic regions (Table 2<sup>77–89</sup>), likely also affected by notable variations in its definition over time and in epidemiological studies. Early estimates (2007) suggested the incidence in the United States (defined as LVEF <50% and occurring in the last month of pregnancy or within 5 months postpartum) to be 1 in 4025 deliveries.<sup>77</sup> Subsequently, PPCM prevalence (defined as LVEF <45%) was estimated as 1 in 2066<sup>79</sup> to 1 in 2772 live births.<sup>83</sup> Most recently, data from the National Inpatient Sample covering the years 2010 to 2020 indicate the overall prevalence of PPCM to be 1 in 951 deliveries and of any cardiomyopathy (including PPCM) during pregnancy to be 1 in 552 deliveries.<sup>85</sup> HF is the most common complication (11%) occurring among obstetric patients with known heart disease during pregnancy and in the postpartum period.<sup>90</sup> However, contrary to expectations, data from a nationwide study in France spanning 2010 to 2018 demonstrated that the vast majority (80%) of women who developed HF in the perinatal period had no history of heart disease.<sup>89</sup> Other forms of de novo cardiomyopathy with reduced LVEF observed during the perinatal period include stress-induced cardiomyopathy,<sup>91–93</sup> which meets the same diagnostic criteria for PPCM, and myocarditis-related cardiomyopathy.<sup>61</sup> However, the unique epidemiological features of these other conditions in the context of pregnancy are not well characterized in the literature.

### Disparities in HF-Related Maternal Morbidity and Mortality

There are notable disparities in perinatal HF risk and outcomes in the United States. Data from the US National Inpatient Sample from 2016 to 2018, demonstrated that Black women and Native American women were more frequently diagnosed with PPCM (adjusted odds ratio, 1.89 and 1.60, respectively) compared with White women.<sup>94</sup> However, from 2009 to 2010, Asian women had the highest in-hospital mortality rate at 8.3% (9/108)<sup>95</sup> compared with other racial groups, with 2.8% (49/1721) among Black women and 0.3% (5/1601) among White women. A diagnosis of HF, regardless of disease phenotype, in the perinatal period is associated with a substantial increase in the likelihood of maternal death, with adjusted odds ratios of 15.9 (95% CI, 19.3–52.8), 31.9 (95% CI, 9.5–26.6), and 4.0 (95% CI, 3.1–5.3) during pregnancy, at delivery, and postpartum respectively.<sup>4</sup> According to data from the US Centers for Disease Control Pregnancy Mortality Surveillance System (2007–2016), pregnancy-related mortality attributed to cardiomyopathy was highest among American Indian/Alaska Native women, accounting for 14.5% (17/117) of all deaths.<sup>96,97</sup> It was also identified as the second most common cause of death among Black women at 14.2% (345/2432), and other cardiovascular conditions accounted for the leading cause of death at 16.2% (393/2432).<sup>97</sup> Data from Maternal Mortality Review Committees in 36 US states from 2017 to 2019 showed the same pattern for Black women.<sup>2</sup> Mortality rates for Black women diagnosed with PPCM have also been shown to be 5 times higher than in White women (mortality rate ratio, 4.86 [95% CI, 2.93–8.12]) according to an analysis of US vital statistics data (2016–2017).<sup>98</sup> Unfortunately, Black women with PPCM are also more likely to be diagnosed later and exhibit a lower rate of left ventricular recovery: 57% (69/121) versus 76% (75/99) compared with other racial groups.<sup>99</sup> Data from the US IPAC cohort (2009–2012) showed that the overall mortality rate for PPCM at 12 months was 4.4% (4/91).<sup>100</sup> In the ROPAC registry (2007–2011), based on data from 28 countries, HF-associated maternal mortality was 4.8%.<sup>101</sup> Globally, according to a systematic review of 47 studies, the observed 12-month mortality rates associated with PPCM were reported to be highest in Africa (15.2% [95% CI, 9.8–21.4]) and the Asia/Pacific area (13.6% [95% CI, 2.7–29.8]). These regions also had a lower frequency of patients who achieved LVEF improvement to values  $\geq 50\%$ .<sup>102</sup>

### EVALUATION AND DIAGNOSIS

Delays in the recognition and diagnosis of HF during the perinatal period are significant contributors to adverse maternal outcomes.<sup>30,99</sup> Therefore, early recognition and diagnosis present a crucial opportunity to improve

**Table 2. Epidemiology of Perinatal HF: Comparative Incidence and Prevalence Across US Racial and Ethnic Groups and International Populations**

| Estimate   | Definition                          | United States                                                                                                                                                                                                         | Global                                                                                                                                                                                                                                                   |
|------------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Incidence  | PPCM defined as LVEF <50%           | Southern California, <sup>77</sup> 2007:<br>Overall: 1 in 4025<br>White: 1 in 4075<br>Black: 1 in 1421<br>Hispanic: 1 in 9861<br>Asian: 1 in 2675                                                                     | Japan, <sup>78</sup> 2011:<br>1 in 20 000                                                                                                                                                                                                                |
| Incidence  | PPCM defined as LVEF <45%           | Northern California, <sup>79</sup> 2011:<br>Overall: 1 in 2066<br>White: 1 in 2451<br>Black: 1 in 662<br>Hispanic: 1 in 6711<br>Chinese: 1 in 4525<br>Filipino: 1 in 978<br>Other A/PI: 1 in 2571<br>Other: 1 in 1938 | Nigeria, <sup>80</sup> 2020:<br>Northwest: 1 in 96–117<br>Northeast: 1 in 177–340<br>North-Central: 1 in 2700<br>Southwest: 1 in 1170<br>South-South: 1 in 1350<br>Haiti, <sup>81</sup> 2005:<br>1 in 300<br>Denmark, <sup>82</sup> 2017:<br>1 in 10 149 |
| Prevalence | PPCM defined as LVEF <45%           | North Carolina, <sup>83</sup> 2012:<br>Overall: 1 in 2772<br>White: 1 in 4266<br>Black: 1 in 1087<br>Hispanic: 1 in 31 140<br>Asian: 1 in 5954                                                                        | Israel, <sup>84</sup> 2018:<br>1 in 3239                                                                                                                                                                                                                 |
| Prevalence | PPCM defined by ICD codes           | National Inpatient Sample, <sup>85</sup> 2025:<br>Overall: 1 in 951<br>White: 1 in 1289<br>Black: 1 in 326<br>Hispanic: 1 in 1946<br>ANHOPI: 1 in 5954<br>AI/AN: 1 in 646                                             | South Korea, <sup>86</sup> 2018:<br>1 in 1741                                                                                                                                                                                                            |
| Incidence  | Peripartum HF* defined by ICD codes |                                                                                                                                                                                                                       | Sweden, <sup>87</sup> 2017:<br>1 in 5719<br>Finland, <sup>88</sup> 2024:<br>1 in 8696<br>France, <sup>89</sup> 2025:<br>1 in 2808                                                                                                                        |

A/PI indicates Other Asian or Pacific Islander; AI/AN, American Indian/Alaska Native; ANHOPI, Asian, Native Hawaiian, and Other Pacific Islander; HF, heart failure; ICD, *International Classification of Diseases*; LVEF, left ventricular ejection fraction; and PPCM, peripartum cardiomyopathy.

\*Peripartum HF definition varied across studies and included HF and cardiomyopathy identified in the last 3 months of pregnancy through 6 months postpartum (Sweden), all forms of HF or cardiomyopathy at any time during pregnancy through 6 months postpartum (Finland), and HF hospitalization at any time during pregnancy through 6 weeks postpartum (France).

maternal health outcomes by facilitating timely and appropriate treatment interventions.

## HF Screening and Risk Assessment During Pregnancy

Current risk assessment recommendations are tailored mostly for women with preexisting CVD and begin with

preconception counseling and evaluation.<sup>76</sup> Widely used obstetric risk assessment tools such as the CARPREG II (Cardiac disease in Pregnancy Study) risk score or the modified World Health Organization 2.0 classification<sup>15b</sup> of maternal cardiovascular risk provide estimated risk for adverse cardiac events rather than risk for HF or LVSD. Of note, 39% (106/307) of adverse cardiac events in the CARPREG II study were from left-sided HF.<sup>9</sup> More

recently, the American Heart Association's PREVENT equation provides an estimated risk of HF for adults 30 to 79 years of age.<sup>103,104</sup> However, the utility of the PREVENT equation for de novo HF risk prediction has not been evaluated in obstetric populations.

The only HF screening tool that exists for use among pregnant and postpartum women in general is the CMQCC (California Maternal Quality Care Collaborative) CVD Risk Assessment Algorithm,<sup>33</sup> developed in 2017 and endorsed by the American College of Obstetrics and Gynecology.<sup>30</sup> This risk assessment algorithm uses a combination of symptoms (specific for HF) along with vital signs, risk factors, and physical examination findings in evaluating HF risk. The initial diagnostic tests recommended include an ECG, natriuretic peptides, and an echocardiogram. One prospective validation study demonstrated this tool to have a positive predictive value of 34% (12/35) for any CVD among those with positive screens who had complete follow-up.<sup>105</sup> Patients identified as high risk with the CMQCC screening tool were about twice as likely (odds ratio, 1.69 [95% CI, 1.53–1.87]) to have an abnormal finding on subsequent cardiovascular diagnostic testing compared with patients who underwent testing based solely on clinical judgment.<sup>106</sup>

### Recognition and Diagnosis in the Postpartum Period

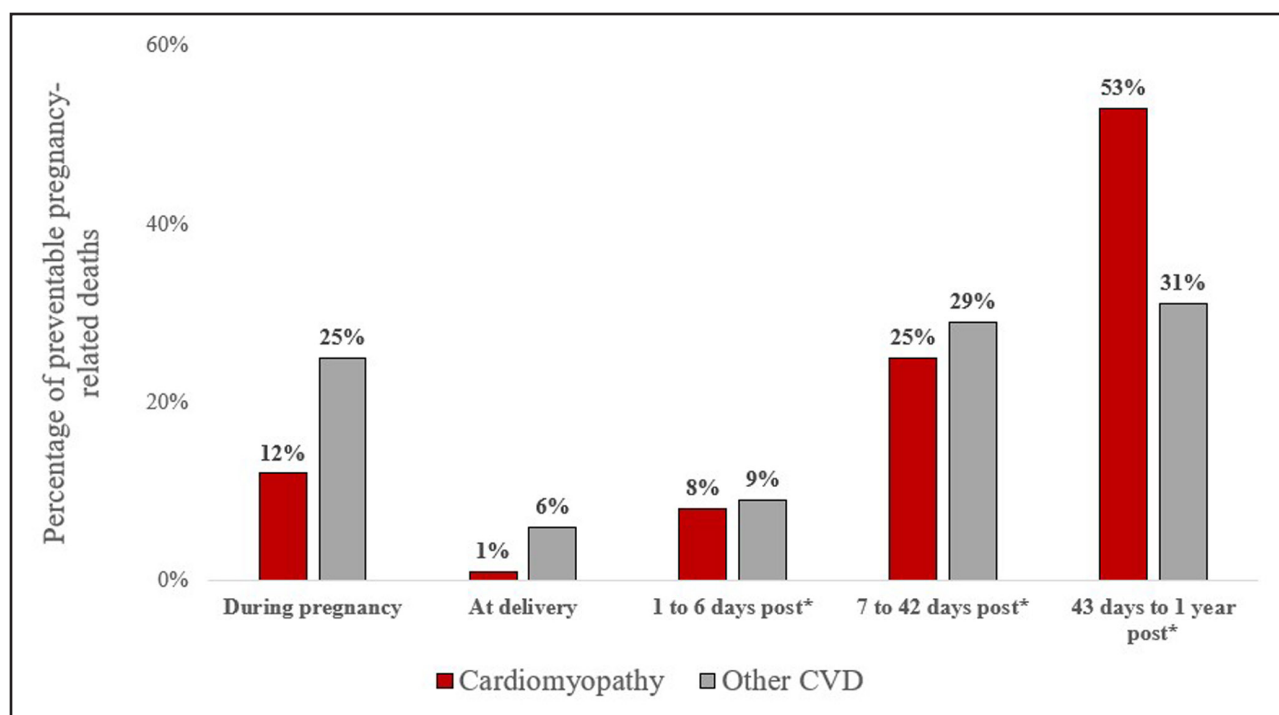
Cardiovascular conditions, particularly cardiomyopathy, are leading contributors to maternal death in the late postpartum period,<sup>107–109</sup> which extends beyond 6 weeks and up to 12 months after delivery. In fact, 78% of preventable pregnancy-related deaths attributable to cardiomyopathy occurred between 7 days and 12 months after the end of pregnancy<sup>108</sup> (Figure 1). Of all cardiomyopathy-related deaths, PPCM accounted for 56%, hypertrophic cardiomyopathy accounted for 7.1%, and other cardiomyopathies accounted for 36.9%.<sup>108</sup> Therefore, prompt identification of HF in the postpartum window is critically important to reduce adverse maternal outcomes.<sup>10</sup> Optimal postpartum care considerations should ideally include long-term follow-up care that extends beyond the traditional 6-week postpartum period and earlier than 6 weeks for at-risk patients.<sup>110</sup> This includes women with preexisting CVD at moderate to high risk assessed with either the CARPREG II ( $\geq 2$  points) or modified World Health Organization 2.0 (class II–III or higher) tools, those with a positive CMQCC screen (at least 1 point in each group or a total of  $\geq 4$  points), and those with other independent risk factors such as hypertensive disorders of pregnancy, multiple gestation, older maternal age ( $>40$  years), use of assisted reproductive technology, dilated cardiomyopathy genetic variants, and prolonged use of tocolytic agents, as well as those who develop new HF symptoms in the postpartum period.<sup>15b,111</sup> Monitoring may involve home visits and alternatives to in-person clinic appointments

such as telemedicine or other virtual or hybrid follow-up options,<sup>110,112,113</sup> in addition to leveraging digital technologies for remote monitoring and serial symptom assessments.<sup>110</sup> Biometric signal, sensor, and patient-reported outcomes data gathered through multiple touchpoints in the postpartum window, when women are often lost to care, may provide an opportunity for early recognition of actionable clinical issues and the initiation of appropriate clinical interventions.

## MANAGEMENT

### Preconception Cardiovascular Care for Women With Preexisting HF

Among women of reproductive age with preexisting LVSD or those identified as being at high risk for HF, cardiovascular care should ideally begin before conception with risk assessment and counseling and continue through the postpartum period.<sup>114</sup> A structured framework for cardiovascular risk assessment and diagnostic testing before pregnancy is summarized in [Supplemental Table 1a](#).<sup>15a,16,24,33,47,60,63,76,114–123</sup> Initial assessment should include a comprehensive medical history and physical examination, a review and optimization of pharmacological therapy (Table 3<sup>15a,124–133</sup> and Figure 2) to improve fetal safety profile, an assessment of maternal cardiovascular risk, and patient education. Suggested clinical tests, include an ECG, resting transthoracic echocardiography, serum natriuretic peptide levels (at initial assessment and serially through pregnancy as appropriate), and stress echocardiography, can be considered to assess myocardial reserve before pregnancy ([Supplemental Table 1a](#)<sup>15a,16,24,33,47,60,63,76,114–123</sup> and Figure 2). The engagement of a multidisciplinary cardio-obstetric team is important to ensure optimal care.<sup>25,112</sup> Pregnancy-associated risk varies among patients with HF and is influenced by multiple factors, including the specific pathology, symptom severity, and degree of left ventricular dysfunction. Existing risk stratification tools include the modified World Health Organization 2.0<sup>15b,134</sup> classification of maternal cardiovascular risk and the CARPREG II risk score.<sup>9,135</sup> According to both tools, women with LVSD, whether preexisting or de novo, are classified as being at moderate to high risk for maternal morbidity and mortality. Specific characteristics associated with the highest risk during pregnancy are LVEF  $<30\%$ , New York Heart Association class III or IV symptoms, and a history of PPCM with persistent LV dysfunction.<sup>114</sup> Contraception counseling should be centered on patient preferences and include a discussion of the risks and benefits associated with various contraceptive methods. In this context, long-acting reversible contraceptives, specifically hormonal intrauterine devices, are the preferred method of contraception<sup>76,136</sup>; in addition, subdermal implants (also a long-acting reversible contraceptive method) may be



**Figure 1. Timing of preventable pregnancy-related deaths for cardiomyopathy vs other cardiovascular conditions.**

This bar graph illustrates the distribution of preventable pregnancy-related deaths attributed to cardiomyopathy compared with other cardiovascular conditions (excludes hypertensive disorders of pregnancy and cerebrovascular accidents), categorized by timing: during pregnancy, labor and delivery, and across the postpregnancy period. The figure highlights that a substantial proportion of deaths attributable to cardiomyopathy occur in the postpregnancy window (7 days–12 months). These data are based on 210 pregnancy-related deaths attributed to cardiovascular conditions, among which 84 were due to cardiomyopathy and 126 to other cardiovascular conditions. CVD indicates cardiovascular disease. \*Post refers to the end of pregnancy, that is, after childbirth or pregnancy loss. Adapted from Briller et al.<sup>108</sup> Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an Open Access Article under the CC BY License. (<http://creativecommons.org/licenses/by/4.0/>).

considered<sup>136,137</sup> (Supplemental Table 2). It is important to note that estrogen-containing methods are not recommended for women with moderate or severe LV dysfunction given the increased risk for thrombosis. Detailed contraceptive considerations, including risks and side effects, among women with perinatal HF and cardiomyopathies have been published.<sup>76,138</sup>

### Management of Preexisting LVSD and Chronic HF During Pregnancy

Women with preexisting LVSD and chronic HF should have guideline-directed medical therapy adjusted to a pregnancy safe regimen (Table 3 and Figure 2) as soon as pregnancy is confirmed<sup>15a</sup> if the patient was not evaluated before pregnancy. Generally,  $\beta$ -blockers, nitrates, hydralazine, and loop diuretics are considered safe in pregnancy and should be continued.<sup>114</sup> A baseline transthoracic echocardiogram along with natriuretic peptides can help guide prenatal care, referrals, and appropriate interventions. Hemodynamic changes vary with each trimester; therefore, serial transthoracic echocardiograms in each trimester should be considered<sup>114</sup> (Supplemental Table 1b<sup>15a,16,24,33,47,60,63,76,114–123</sup> and Figure 2). A new

change in symptoms, clinical status, or medication requirement should trigger a repeat clinical assessment, including natriuretic peptides and an echocardiogram.<sup>15a</sup> A detailed overview of cardiovascular risk assessment and diagnostic testing during pregnancy is provided in Supplemental Table 1b.<sup>15a,16,24,33,47,60,63,76,114–123</sup>

### Acute HF Management in Pregnancy and Postpartum

The key steps in acute HF management are to stabilize maternal hemodynamics, ascertain the cause of HF, determine disease severity, evaluate the condition of the fetus, ensure fetal and uterine perfusion, identify complications, and initiate appropriate treatment.<sup>8</sup> Common complications that may co-occur with acute HF presentations in the perinatal period include arrhythmias, cardiogenic shock, thromboembolism,<sup>15a</sup> and uteroplacental hypoperfusion.<sup>139</sup> Management strategies and therapeutic interventions for acute HF during pregnancy and the postpartum period have previously been published.<sup>8,76,114</sup> The primary goals of care include decongestion with loop diuretics, afterload reduction to improve contractility and decrease mortality,



**Table 3. Oral Pharmacological Therapies for HF With LVSD During Pregnancy and in the Postpartum Period**

| Medication category | Medication examples                                     | Can be used during pregnancy? | What to do if patient is planning a pregnancy or becomes pregnant on this therapy | Can be used during lactation?                                             | Can be used if postpartum and not breastfeeding? | Starting/maximal oral dose                                                                                   | Common side effects and disadvantages                                                                                                                     |
|---------------------|---------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| ACE inhibitors      |                                                         |                               |                                                                                   |                                                                           |                                                  |                                                                                                              |                                                                                                                                                           |
|                     | Captopril                                               | Not recommended               | Discontinue                                                                       | Yes. Excreted in breastmilk, but infant risk is minimal.                  | Yes                                              | Start at 6.25 mg 3 times daily; maximum 50 mg 3 times daily                                                  | Multiple daily dosing. Hypotension, hyperkalemia, cough, dysgeusia                                                                                        |
|                     | Enalapril†                                              | Not recommended               | Discontinue                                                                       | Yes. Excreted in breastmilk, but infant risk is minimal.                  | Yes                                              | Start at 2.5 mg twice daily; maximum 20 mg twice daily                                                       | Hyperkalemia                                                                                                                                              |
|                     | Lisinopril                                              | Not recommended               | Discontinue                                                                       | Not recommended. Limited data suggest that infant risk is likely minimal. | Yes                                              | Start at 2.5 mg daily; maximum 40 mg daily                                                                   | Hypotension, cough                                                                                                                                        |
| ARBs                |                                                         |                               |                                                                                   |                                                                           |                                                  |                                                                                                              |                                                                                                                                                           |
|                     | Losartan                                                | Not recommended               | Discontinue                                                                       | Not recommended. No data                                                  | Yes                                              | Start at 25 mg daily; maximum 100–150 mg daily                                                               | Hypotension, hyperkalemia                                                                                                                                 |
|                     | Valsartan                                               | Not recommended               | Discontinue                                                                       | Not recommended. Limited data                                             | Yes                                              | Start at 20–40 mg daily; maximum 160 mg twice daily                                                          | Hypotension, elevated BUN or Cr, cough                                                                                                                    |
| ARNIs               |                                                         |                               |                                                                                   |                                                                           |                                                  |                                                                                                              |                                                                                                                                                           |
|                     | Sacubitril/valsartan                                    | Not recommended               | Discontinue                                                                       | Not recommended. Limited data suggest that infant risk is likely minimal. | Yes                                              | Start at sacubitril 24 mg/valsartan 26 mg twice daily; maximum sacubitril 97 mg/valsartan 103 mg twice daily | Hypotension, hyperkalemia                                                                                                                                 |
| β-Blockers          |                                                         |                               |                                                                                   |                                                                           |                                                  |                                                                                                              |                                                                                                                                                           |
|                     | Metoprolol tartrate (not FDA approved for HF treatment) | Yes*                          | Continue                                                                          | Yes. Infant risk is minimal.                                              | Yes                                              | Start at 25 mg twice daily; maximum 100 mg twice daily                                                       | Bradycardia, hypotension. Low potential risk for fetal growth restriction                                                                                 |
|                     | Metoprolol Succinate†                                   | Yes*                          | Continue                                                                          | Yes. Infant risk is minimal.                                              | Yes                                              | Start at 12.5–25 mg daily; maximum 200 mg daily                                                              | Bradycardia, hypotension. Low potential risk for fetal growth restriction                                                                                 |
|                     | Carvedilol                                              | Yes* Not preferred            | May continue. Consider switching to metoprolol                                    | Generally, yes. Limited data with carvedilol compared with metoprolol     | Yes                                              | Start at 3.125 mg twice daily; maximum 25 mg twice daily                                                     | Bradycardia, hypotension, hyperglycemia, weight gain                                                                                                      |
|                     | Bisoprolol                                              | Limited data                  | May continue. Consider switching to metoprolol                                    | Yes, limited data                                                         | Yes                                              | Start at 1.25 mg daily; maximum 10 mg/d                                                                      | Diarrhea, headache                                                                                                                                        |
| Diuretics           |                                                         |                               |                                                                                   |                                                                           |                                                  |                                                                                                              |                                                                                                                                                           |
| Loop diuretics      |                                                         |                               |                                                                                   |                                                                           |                                                  |                                                                                                              |                                                                                                                                                           |
|                     | Furosemide†                                             | Yes                           | Continue if evidence of volume overload persists. Consider dose adjustment        | Yes. Excreted in breastmilk, but infant risk is minimal                   | Yes                                              | Start at 20 mg daily or twice daily; maximum 600 mg total/d                                                  | Photosensitivity, hypokalemia, hypomagnesemia. Can suppress lactation. Associated fetal risks include oligohydramnios and fetal electrolyte abnormalities |

(Continued)

**Table 3. Continued**

| Medication category                   | Medication examples  | Can be used during pregnancy?                                              | What to do if patient is planning a pregnancy or becomes pregnant on this therapy | Can be used during lactation?                                       | Can be used if postpartum and not breastfeeding? | Starting/maximal oral dose                                                                                                  | Common side effects and disadvantages                                                                                                     |
|---------------------------------------|----------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | Bumetanide           | Yes                                                                        | Continue if evidence of volume overload persists. Consider dose adjustment        | Generally, yes. Limited data on bumetanide compared with furosemide | Yes                                              | Start at 0.5–1 mg daily or twice daily; maximum 10 mg daily                                                                 | Hypotension, hypokalemia, hyperuricemia, hyponatremia. Associated fetal risks include oligohydramnios and fetal electrolyte abnormalities |
|                                       | Torsemide            | Not recommended. Limited data                                              | Discontinue. Limited data                                                         | Not recommended. No data                                            | Yes                                              | Start at 10–20 mg daily; maximum 200 mg daily                                                                               | Polyuria, hypotension                                                                                                                     |
| Thiazide diuretics                    |                      | May be used as adjunct diuretic if insufficient response to loop diuretics |                                                                                   |                                                                     |                                                  |                                                                                                                             |                                                                                                                                           |
|                                       | Chlorthalidone       | No data                                                                    | Discontinue                                                                       | No data                                                             | Yes                                              | Start at 12.5–25 mg daily; maximum 100 mg/d                                                                                 | Hyperuricemia, electrolyte abnormalities. Can suppress lactation                                                                          |
|                                       | Chlorothiazide       | Not recommended.                                                           | Discontinue                                                                       | May be used. Limited data                                           | Yes                                              | Start at 250–500 mg daily or twice daily; maximum 1000 mg/d                                                                 | Photosensitivity, hyperglycemia, hyperuricemia, electrolyte abnormalities                                                                 |
|                                       | Hydrochlorothiazide  | Limited data                                                               | Consider discontinuation. May resume later if needed.                             | Excreted in breastmilk. Limited data. Consider alternative.         | Yes                                              | Start at 12.5–25 mg daily or twice daily; maximum 200 mg/d                                                                  | Hypotension, phototoxicity, electrolyte abnormalities                                                                                     |
|                                       | Metolazone           | Not recommended. Limited data                                              | Discontinue                                                                       | No data                                                             | Yes                                              | Start at 2.5 mg daily; maximum 20 mg/d                                                                                      | Hypotension, electrolyte abnormalities, hyperuricemia                                                                                     |
| Mineralocorticoid receptor antagonist |                      |                                                                            |                                                                                   |                                                                     |                                                  |                                                                                                                             | Hyperkalemia                                                                                                                              |
|                                       | Spironolactone†      | Not recommended                                                            | Discontinue                                                                       | Yes. Excreted in breastmilk, but infant risk is minimal.            | Yes                                              | Start at 12.5–25 mg daily; maximum 25–50 mg/d. For eGFR 31–49 mL/min/1.73 m <sup>2</sup> , dosing should be reduced by half | Diarrhea, nausea/vomiting, electrolyte abnormalities                                                                                      |
|                                       | Eplerenone           | Not recommended. Limited data                                              | Discontinue                                                                       | Limited to no data available on safety                              | Yes                                              | Start at 25 mg daily; maximum 50 mg/d. For eGFR 31–49 mL/min/1.73 m <sup>2</sup> , dosing should be reduced by half         | Increased serum Cr, hyperkalemia, hepatic enzyme abnormalities                                                                            |
| SGLT2 inhibitors                      |                      |                                                                            |                                                                                   |                                                                     |                                                  |                                                                                                                             |                                                                                                                                           |
|                                       | Dapagliflozin        | Not recommended. Limited data                                              | Discontinue                                                                       | No data                                                             | Yes                                              | Start at 10 mg daily; maximum 10 mg/d                                                                                       | UTI, female genital infection, nasopharyngitis, hypotension                                                                               |
|                                       | Empagliflozin        | Not recommended. Limited data                                              | Discontinue                                                                       | No data                                                             | Yes                                              | Start at 10 mg daily; maximum 10 mg/d                                                                                       | UTI, female genital infection                                                                                                             |
| Vasodilators                          |                      |                                                                            |                                                                                   |                                                                     |                                                  |                                                                                                                             |                                                                                                                                           |
|                                       | Hydralazine          | Yes. Limited data                                                          | Continue                                                                          | Yes. Infant risk is minimal                                         | Yes                                              | Start at 25–50 mg 3–4 times daily; maximum 200–300 mg/d                                                                     | Edema, palpitations, tachycardia, headache, gastrointestinal upset                                                                        |
|                                       | Isosorbide dinitrate | Yes. Limited data                                                          | Continue                                                                          | No data. Caution advised                                            | Yes                                              | Start at 20–30 mg 3–4 times daily; maximum 120 mg/d                                                                         | Hypotension, headache                                                                                                                     |

(Continued)

**Table 3. Continued**

| Medication category | Medication examples                                                       | Can be used during pregnancy?                                                                                                                                                                                                                            | What to do if patient is planning a pregnancy or becomes pregnant on this therapy   | Can be used during lactation?                                               | Can be used if postpartum and not breastfeeding? | Starting/maximal oral dose                                                                                                                | Common side effects and disadvantages                                                                         |
|---------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
|                     | Fixed-dose combination isosorbide dinitrate and hydralazine               | Limited data                                                                                                                                                                                                                                             | May continue if needed                                                              | Limited data                                                                | Yes                                              | Start at isosorbide dinitrate 20 mg/hydralazine 37.5 mg 3 times daily; maximum isosorbide dinitrate 40 mg/hydralazine 75 mg 3 times daily | Nausea, dizziness, headache                                                                                   |
| Cardiac glycoside   |                                                                           |                                                                                                                                                                                                                                                          |                                                                                     |                                                                             |                                                  |                                                                                                                                           |                                                                                                               |
|                     | Digoxin (in symptomatic patients despite GDMT or unable to tolerate GDMT) | Can be used with caution. Limited data                                                                                                                                                                                                                   | May continue after weighing benefits and risks                                      | Yes. Infant risk is minimal                                                 | Yes                                              | Start at 0.125-0.25 mg daily; maximum individualized to achieve serum concentration 0.5–0.9 ng/mL                                         | Nausea, vomiting, headache, mental health effects (anxiety, depression, hallucinations, delirium)             |
| Anticoagulants†     |                                                                           |                                                                                                                                                                                                                                                          |                                                                                     |                                                                             |                                                  |                                                                                                                                           |                                                                                                               |
|                     | Warfarin (therapeutic and prophylactic)                                   | May be used after risk-benefit discussion with patient in the 1st trimester (at a dose ≤5 mg/d) and in the 2nd and 3rd trimesters, especially if patient has a mechanical valve or LVAD<br><br>Subcutaneous LMWH† is preferred for all other indications | May continue after weighing benefits and risks and considering current patient dose | Yes. Infant risk is minimal with dose ≤12 mg/d.                             | Yes                                              | Start at 2.5–5 mg daily and titrated to a target INR of 2 to 3                                                                            | Alopecia, bleeding.<br><br>Associated fetal risk of embryopathy (including limb defects and nasal hypoplasia) |
|                     | Rivaroxaban (prophylactic)                                                | Not recommended. Limited data                                                                                                                                                                                                                            | Discontinue                                                                         | Limited data. Minimally excreted in breastmilk, but infant risk is minimal. | Yes                                              | Start at 10 mg once daily; maximum 10 mg/d                                                                                                | Bleeding, gastroenteritis, vomiting, cough                                                                    |
|                     | Apixaban (prophylactic)                                                   | Not recommended. Limited data                                                                                                                                                                                                                            | Discontinue                                                                         | Not recommended. Limited data                                               | Yes                                              | Start at 2.5–5 mg twice daily; maximum 5 mg twice daily                                                                                   | Bleeding                                                                                                      |

ACE indicates angiotensin-converting enzyme; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor neprilysin inhibitor; BUN, blood urea nitrogen; Cr, creatinine; eGFR, estimated glomerular filtration rate; FDA, US Food and Drug Administration; GDMT, guideline-directed medical therapy; INR, international normalized ratio; LMWH, low-molecular-weight heparin; LVAD, left ventricular assist device; LVSD, left ventricular systolic dysfunction; SGLT2, sodium-glucose cotransporter-2; and UTI, urinary tract infection.

Relevant references: <sup>15a</sup> and <sup>124</sup> through <sup>133</sup>.

Dosing recommendations for LMWH have been published elsewhere.<sup>130</sup>

\*Potential risk for fetal growth restriction with long term use of metoprolol and carvedilol.

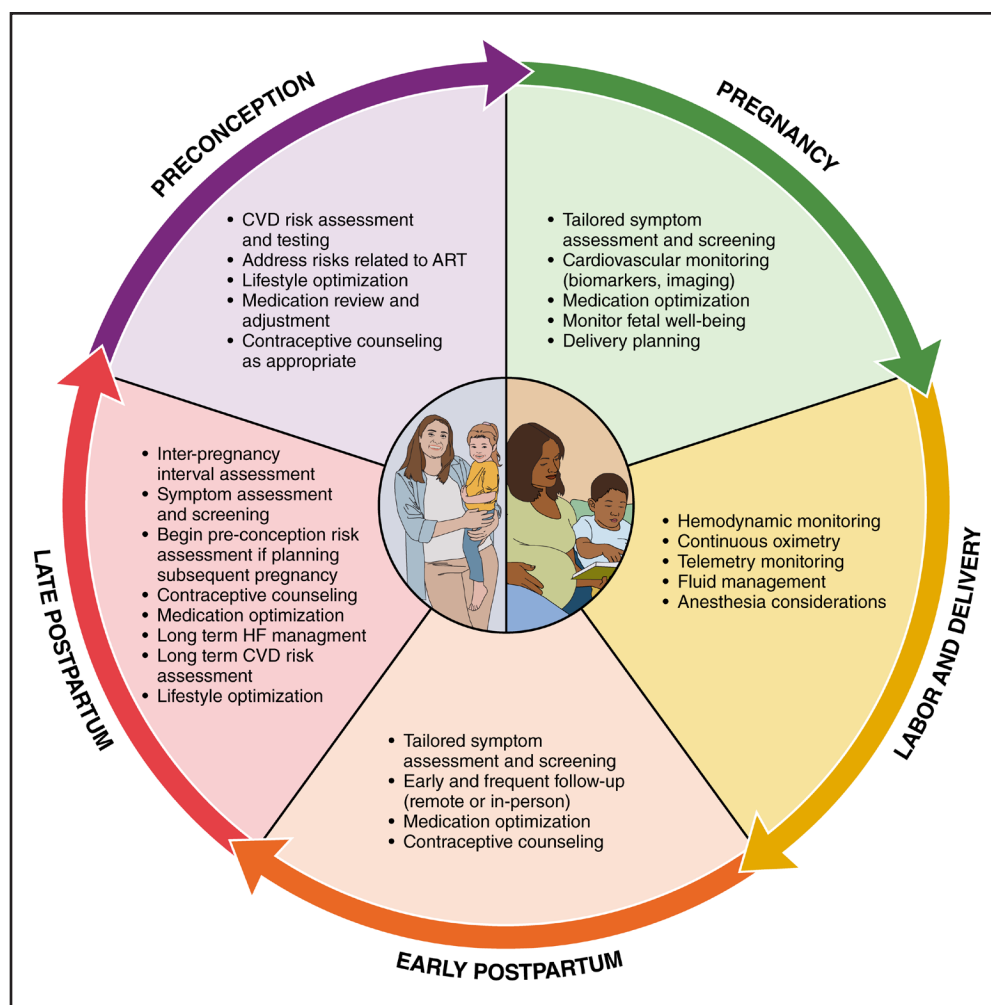
†Preferred option or first line.

‡Adjunctive therapy if LVEF <30% at PPCM diagnosis and through 8 weeks postpartum.

Sources: Published documents, Micromedex, LactMed.

and  $\beta$ -blockers (if there are no clinical contraindications such as hemodynamic instability or patients requiring inotropic support) to improve left ventricular function and decrease risk of death or hospitalization.<sup>15a</sup> Treatment escalation to advanced HF therapies is usually guided by disease severity, determined by the degree of LVSD and hemodynamic instability.<sup>23,72</sup> Patients

with acute severe HF who develop cardiogenic shock requiring inotropic or mechanical circulatory support should be cared for in centers where specialized cardiovascular expertise (interventional cardiology, advanced HF/transplant cardiology, cardiothoracic surgery, and cardiothoracic anesthesiology) is available<sup>140</sup> in conjunction with other members of the obstetric



**Figure 2. Managing heart failure in the perinatal period: opportunities across the pregnancy-postpartum continuum.**

This figure outlines key phases in the pregnancy-postpartum continuum in which targeted interventions can improve the recognition, management, and outcomes of heart failure in obstetric patients. The connecting arrows shown in the figure are intended to convey that although specific time points are highlighted, care is provided as a continuous process throughout the perinatal period. The points presented represent key considerations, with the recognition that this list is not exhaustive. ART indicates assisted reproductive technology; CVD, cardiovascular disease; and HF, heart failure.

care team. In relation to delivery considerations, the preferred mode of delivery is vaginal when possible because of reduced hemodynamic stress<sup>141</sup> and a lower risk of perioperative complications and maternal morbidity.<sup>140,142</sup> Ideal candidates for trials of labor and the need for assisted delivery (with either forceps or vacuum) should be determined by a multidisciplinary cardio-obstetric team, and planned cesarean deliveries should be reserved for patients with fetal, obstetric, and other maternal indications. Effective neuraxial labor analgesia can reduce the catecholamine surge associated with labor pains, attenuating heart rate and blood pressure fluctuations during labor that may affect hemodynamic stability.<sup>140</sup> Epidural or combined spinal-epidural techniques are also ideal for patients with LVSD because they allow controlled incremental dosing with concomitant titration of vasoactive therapies when needed. Suggested dosing and obstetric

considerations for vasoactive medication use in the perinatal period are reviewed elsewhere.<sup>142</sup> For patients with LVSD undergoing cesarean delivery, neuraxial anesthesia is also preferred because it avoids the hemodynamic stress due to myocardial depression and rapid shifts in preload and afterload associated with induction, intubation, and emergence seen with general anesthesia.<sup>142,143</sup> During labor and delivery, pulse oximetry and hemodynamic and telemetry monitoring should be considered (Figure 2), weighing the potential benefits against the associated discomfort and risks.<sup>115,144</sup> The immediate postpartum period is a vulnerable period for cardiac decompensation in patients with HF because of iatrogenic fluid administration, hemodynamic fluid shifts, and autotransfusion of up to 500 mL uteroplacental blood, leading to increased cardiac preload.<sup>115,145,146</sup> Close evaluation of hemodynamic status, prompt initiation of diuresis,



and early optimization of guideline-directed medical therapy are considered ideal management strategies.

### Postpartum and Follow-Up Care for HF

Following the diagnosis of HF, establishing long-term cardiovascular care is essential. [Supplemental Table 1c](#)<sup>15a,16,24,33,47,60,63,76,114–123</sup> provides a summary of cardiovascular risk assessment and diagnostic testing in the postpartum period. Clinical guideline recommendations align closely with those for nonobstetric patients, with specific exceptions related to pharmacological therapies that are safe during lactation (Table 3 and Figure 2), and the use of anticoagulation in the early postpartum window with LVEF <30%.<sup>15a</sup> In addition, early insertion of implantable cardioverter defibrillators is discouraged among patients diagnosed with PPCM and severely reduced ejection fraction because of a higher likelihood of improvement in LVEF within 3 to 6 months after diagnosis.<sup>124</sup> Wearable cardioverter defibrillators may be considered (with appropriate counseling and education about use) in the interim due to the risk of life-threatening arrhythmias.<sup>15b,76,124</sup> Although there are no specific recommendations on the frequency of care visits in the postpartum window, it has been suggested that follow-up appointments occur earlier than the traditional 6-week follow-up.<sup>4,147</sup> One possible approach is scheduling follow-up appointments at short intervals, every 1 to 2 weeks during the first 6 to 8 weeks postpartum, followed by visits every 12 weeks throughout the first year, and consideration of extended follow-up for up to 3 years.<sup>87</sup> This proposed schedule is intended as a practical framework rather than a formal recommendation. Coordinated handoffs, referrals, and care transitions from obstetric care/maternal-fetal medicine teams to other health care professionals, whether cardiology or primary care, have been identified as essential components of high-quality postpartum care.<sup>110</sup> The role of individual or combinations of remote digital interventions to enhance HF care in the postpartum period remains unknown, although some benefit has been demonstrated in nonobstetric patients with HF after hospital discharge in relation to mortality, hospitalization, quality of life, and self-care behaviors.<sup>148–150</sup> Other successful interventions during the postpartum period have been described, which include the medical home model that facilitates the coordinated scheduling of postpartum and infant care appointments, as well as postpartum transition care clinics designed to deliver specialized cardiovascular care after handoff from obstetric care teams.<sup>151</sup> It is important to recognize that for women who have not yet completed childbearing, each postpartum window represents an interpregnancy interval, and this time frame offers a unique opportunity for preconception risk assessment, counseling, and pregnancy planning (Figure 2).

### NOVEL INSIGHTS, FUTURE DIRECTIONS, AND CONCLUSIONS

A maternal death is a devastating event with far-reaching impact because women often serve as key contributors in both financial and nonfinancial capacities within their households, providing care and support to children, spouses, and elderly relatives. The loss of a mother can adversely affect the survival of her offspring<sup>152,153</sup> and other members of the family, as well as their ability to thrive.<sup>12</sup> Proposed strategies to reduce adverse outcomes associated with perinatal HF include improved risk stratification<sup>154</sup> and early identification of disease.<sup>30</sup> The CMQCC CVD screening algorithm has been successfully implemented into electronic health records of large hospital networks, demonstrating its feasibility in clinical practice.<sup>105,106,155</sup> Additional studies are underway to determine the diagnostic value of this tool in various obstetric care settings.

The use of artificial intelligence has begun to transform many areas of cardiovascular care, with particular emphasis on enhanced detection of ventricular dysfunction.<sup>156</sup> In the pregnant and postpartum patient population, artificial intelligence–powered technologies have demonstrated effectiveness in the detection of pregnancy-related LVSD by leveraging standard 12-lead ECGs<sup>157–161</sup> as well as digital stethoscope ECG and phonocardiogram recordings.<sup>73,161</sup> These artificial intelligence tools have also been evaluated for potential use

**Table 4. Summary of Literature Gaps and Future Directions for Research**

|                                                                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Current epidemiological data on perinatal cardiomyopathy and LVSD                                                                                                                                                                  |
| Development of improved metrics for functional assessment during pregnancy, with a particular focus on patient-reported symptoms and outcomes                                                                                      |
| Establishment of normal echocardiographic reference ranges and natriuretic peptide thresholds in the perinatal period                                                                                                              |
| Development of evidence-based standardized diagnostic protocols for PPCM, accounting for specific gestational windows if appropriate, in addition to comprehensive evaluation protocols to rule out other potential causes of LVSD |
| Creation of clinical guidelines and therapeutic recommendations for the management of hemodynamically unstable HF in pregnancy                                                                                                     |
| Development of tools for early HF risk stratification before conception and in early pregnancy                                                                                                                                     |
| Evaluation and implementation of effective screening tools for detecting cardiomyopathy or LVSD in pregnant and postpartum individuals                                                                                             |
| Evaluation of the efficacy of prophylactic medical therapies in mitigating the risk of HF onset and preventing further decline in left ventricular systolic function during the perinatal period                                   |
| Comprehensive assessment of the safety profile of emerging HF therapies during pregnancy and lactation                                                                                                                             |
| Evaluation of the impact (effectiveness and safety) of structured cardiac rehabilitation on clinical outcomes among women diagnosed with HF during the perinatal period                                                            |

HF indicates heart failure; LVSD, left ventricular systolic dysfunction; and PPCM, peripartum cardiomyopathy.

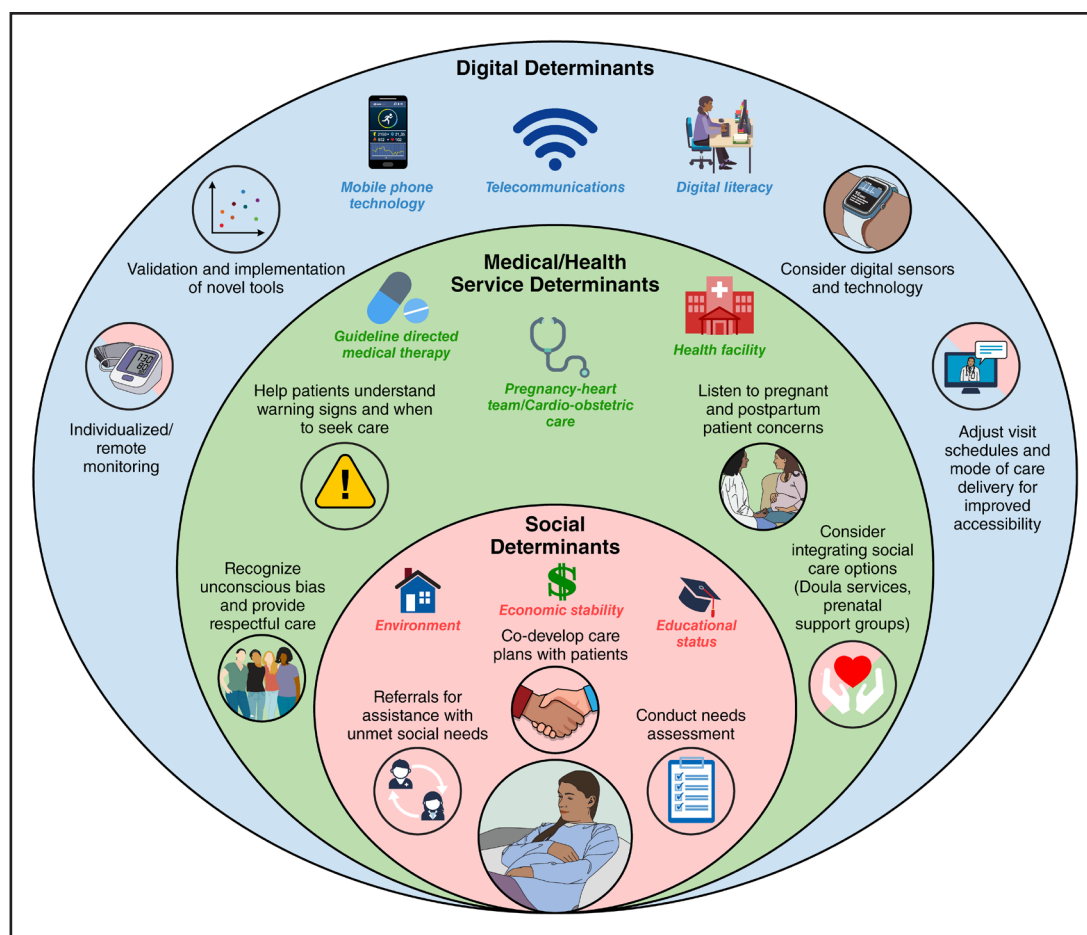
as part of preconception cardiovascular screening with demonstrated effectiveness.<sup>162</sup> With further validation in larger obstetric patient populations, artificial intelligence-guided screening can facilitate enhanced disease detection in clinical and nonclinical environments, early initiation of appropriate pharmacological therapy, clinical monitoring, and large-scale population-based screening to inform epidemiological estimates of LVSD occurrence in obstetric patients.

The ongoing REBIRTH trial (Impact of Bromocriptine on Clinical Outcomes for Peripartum Cardiomyopathy; NCT05180773) is also anticipated to provide important evidence for the use of bromocriptine therapy for the acute and early management of PPCM during the postpartum period. The HOPE trial (Heart Outcomes in Pregnancy Expectations for Mom and Baby; NCT06517628), currently enrolling pregnant individuals with heart disease, aims to offer valuable insights to optimize cardio-obstetrics care and improve perinatal health outcomes.

Despite increasing awareness of perinatal LVSD, differentiating between de novo conditions such as PPCM and preexisting LVSD remains a challenge. Current diag-

nostic imaging modalities and serum biomarkers lack specificity for PPCM. In addition, clinical health records may not sufficiently capture the timing of symptom onset, which may provide some insights into the disease phenotype. It is also important to recognize that LVSD can precede symptom onset, making it an imperfect discriminator on its own. The persistent gaps in knowledge indicate an urgent need for research in the specific areas detailed in Table 4.

Finally, the importance of CVD prevention and addressing social, structural, and digital determinants of health cannot be overemphasized. Achieving optimal cardiovascular health, as outlined by the American Heart Association's Life's Essential 8 metrics (recently including sleep health), is increasingly recognized as important before, during, and between pregnancies.<sup>163</sup> Although pregnancy and the postpartum period are associated with sleep disruption,<sup>164</sup> the direct relationship between sleep quality or sleep disorders and HF development has not been well explored in obstetric populations. Multiple organizations, including the American College of Obstetrics and Gynecology, the Centers for Disease



**Figure 3. Tailored cardio-obstetric care for pregnant and postpartum individuals with and at risk for heart failure.**

This illustration addresses social, digital and health service factors to ensure maternal health equity based on the American College of Obstetricians and Gynecologists's clinical consensus statement on tailored prenatal care delivery, the Center for Disease Control and Prevention's Hear Her Campaign, and the National Institutes of Health Workshop on Technology to Improve Maternal Health.<sup>110,165–167</sup>

CLINICAL STATEMENTS AND GUIDELINES

Control and Prevention, and the National Institutes of Health, have developed clinical guidance documents, health campaigns, educational workshops, and scientific publications emphasizing interventions to improve maternal health equity<sup>110,165–167</sup> (Figure 3). Most recently, the American College of Obstetrics and Gynecology published an expert consensus statement on tailored prenatal care delivery, broadly summarized as (1) conducting a needs assessment, (2) codeveloping care plans with patients, (3) providing referrals for assistance with unmet social needs, (4) adjusting visit schedules and mode of care delivery to improve accessibility, and (5) individualizing health monitoring<sup>165</sup> (Figure 3).

In conclusion, early detection and recognition of HF are crucial to preventing devastating cardiovascular complications and improving maternal outcomes. Ongoing research offers significant promise in both clinical and diagnostic domains, and future efforts should prioritize addressing pending research questions and the identified literature gaps.

ARTICLE INFORMATION

The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a

Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

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|------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------|----------------|--------------------|-------------------------------------------------------------|-----------------------------|
| Demilade A. Adedinsewo | Mayo Clinic Florida                                                    | Mayo Clinic Women’s Health Research Center (research grant Career Development Award)†; Miami Heart Research Institute†   | None                   | None                        | None           | None               | None                                                        | Mayo Clinic (cardiologist)† |
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| Afshan B. Hameed       | University of California, Irvine Medical Center                        | None                                                                                                                     | None                   | None                        | None           | None               | None                                                        | None                        |
| Dennis McNamara        | University of Pittsburgh Medical Center Heart and Vascular Institute   | NHLBI (principal investigator for REBIRTH, a multicenter trial of bro-mocriptine therapy for peripartum cardiomyopathy)† | None                   | None                        | None           | None               | None                                                        | None                        |
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| Karen Florio           | University of Missouri–Columbia                     | NHLBI (HOPE study PI)† | None                   | None                        | None           | None               | None                       | None  |
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